

pH Control Intelligence

*Model-Based Reinforcement Learning
for Chemical Reactor Control*

GRADIENT BOOSTING WM

LOOKAHEAD PLANNER

OFFLINE RL



+3.26×

Target zone
multiplier

+10.1

Reward gain
per step

0.9987

World model
R² accuracy

H=1

Optimal
horizon

Model-Based RL — Three Components, One System

The model does not interact with the reactor during training. It learns, simulates, and plans.

01

What happens next?

World Model

GBR · 200 trees · $R^2=0.9987$ · $MAE=0.028$

Learns the reactor's pH dynamics offline from 35,672 historical transitions. No new reactor interaction required.

$$p\hat{H}(t+1) = f_{\theta}(pH_t, T_t, V_t, buffer, dose)$$

ph_t (88.9%) · $dose_ml$ (10.7%)

02

Which future is best?

Planner

H-step lookahead · $\gamma=0.95$ · H=1 optimal

Evaluates all 5 actions via the world model, computes discounted expected return, selects the best action.

$$a^* = \operatorname{argmax} \sum \gamma^h \cdot r(f_{\theta^h}(st, a), a)$$

Corrects 4.2→7.0 in 6 steps

03

What should I do?

Controller

11.3% in-target · 3.26× vs random

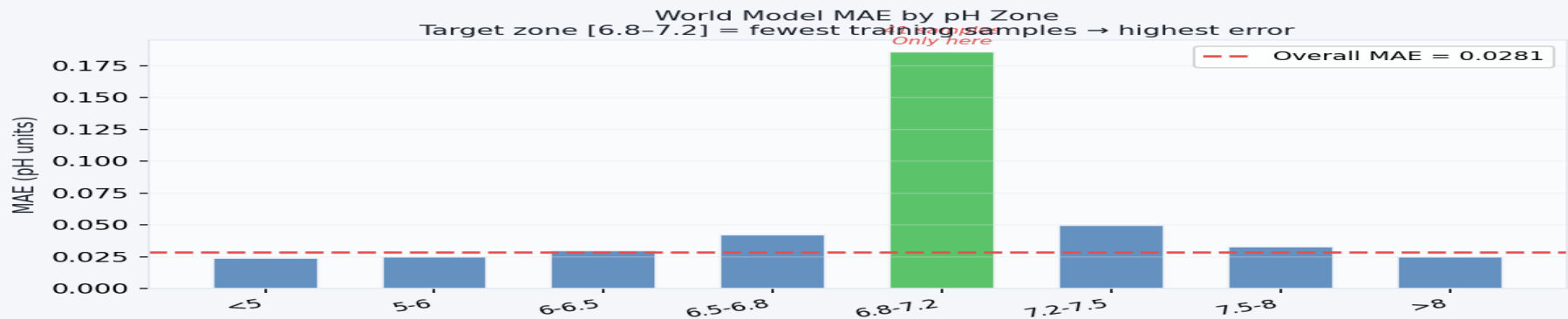
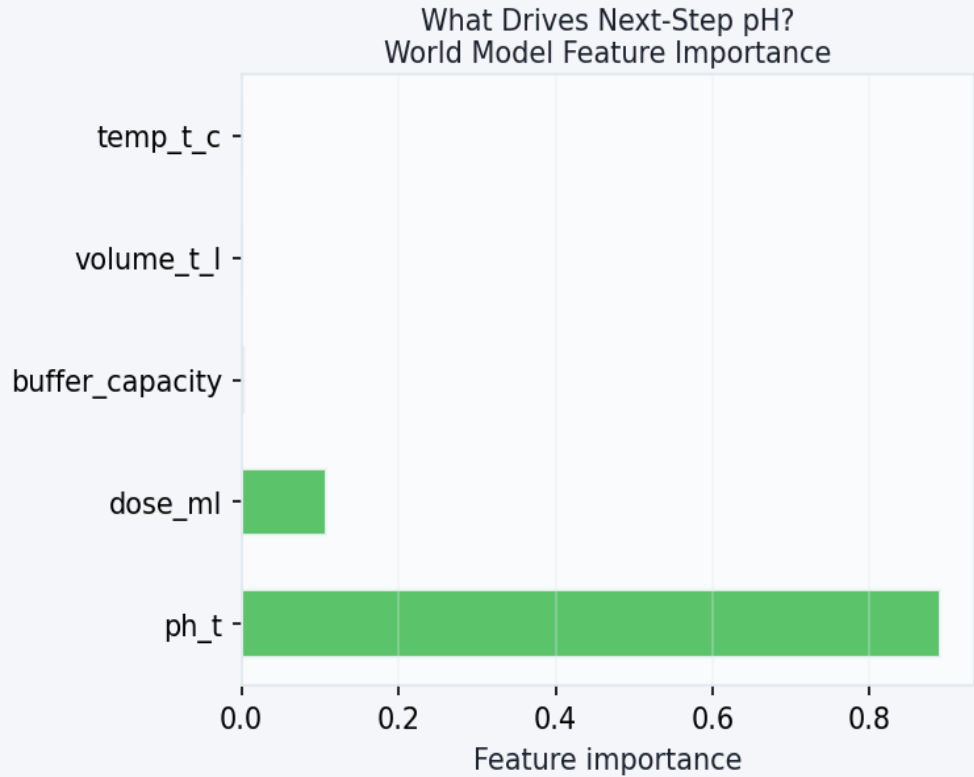
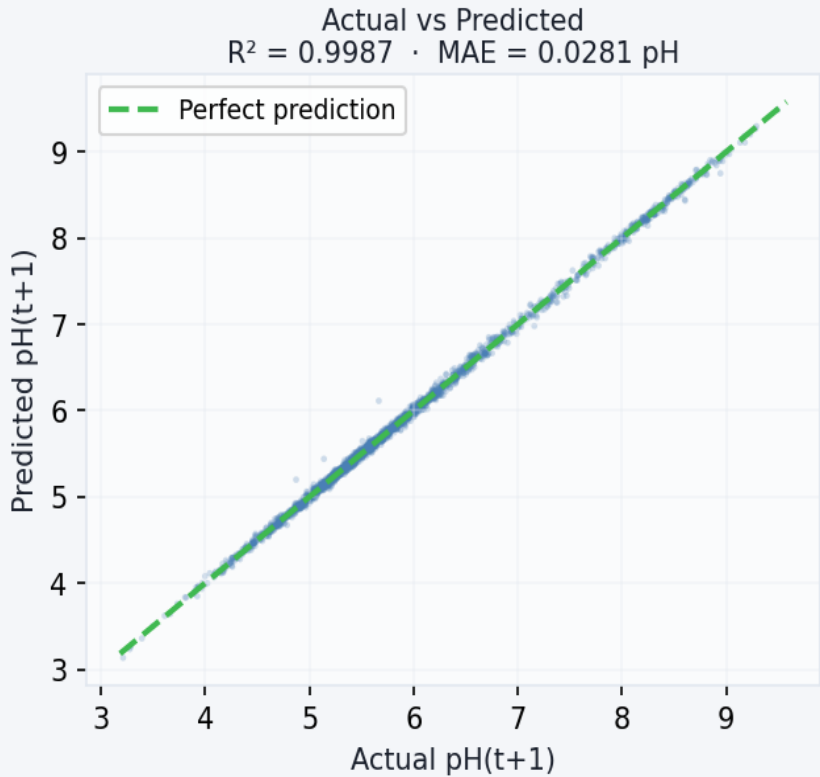
Executes the selected action on the real reactor. Collects the true outcome. Cycles back to the planner.

$$3.5\% \rightarrow 11.3\% \cdot +10.1 \text{ reward/step}$$

$P(MB > random) \rightarrow \text{statistically certain}$

Question 1 — What Happens Next?

The world model learns the reactor's physics from 35,672 offline transitions. No reactor interaction during training.



Target zone [6.8–7.2] has the highest MAE — only 41 training samples. A fundamental offline learning challenge.

$R^2 = 0.9987$

Explains 99.87%
of pH variance

MAE = 0.028

pH units
Sub-threshold

28,537

Training
transitions

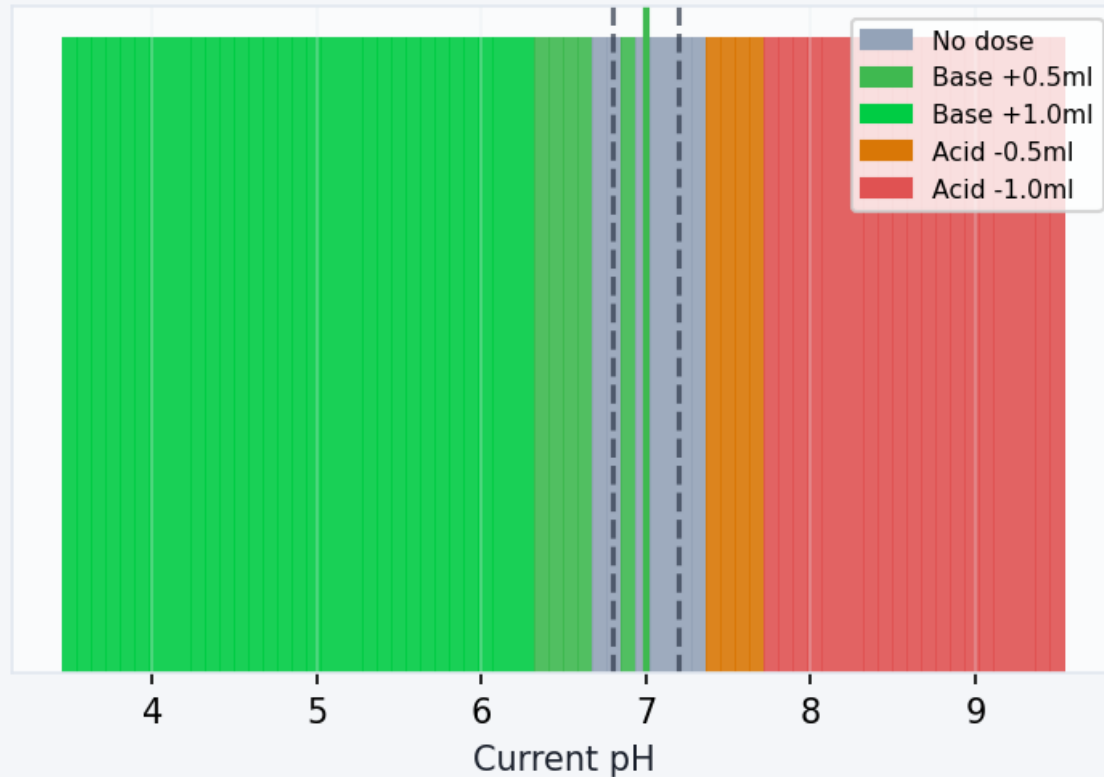
Bias ≈ 0

Symmetric
residuals

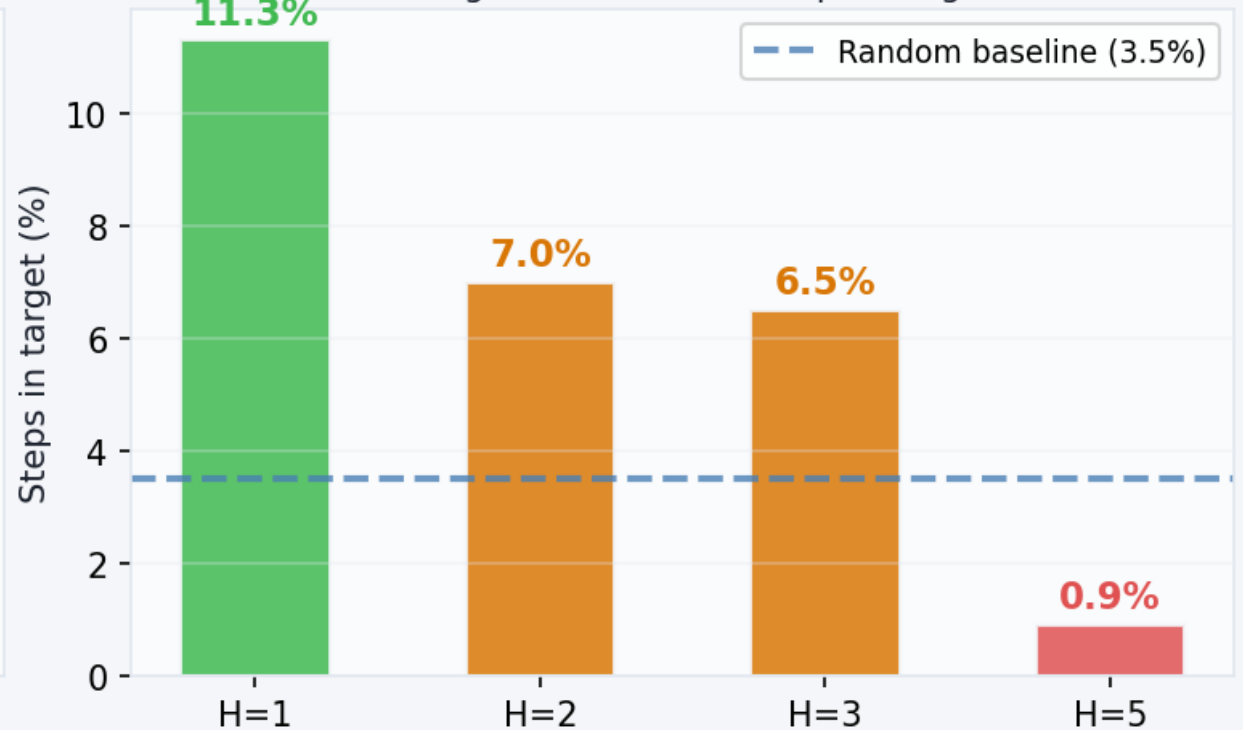
Question 2 — Which Future Is Best?

The planner queries the world model once per action and selects the highest expected return. $H=1$ is optimal.

Planner Action Selection
What the model chooses at each pH level



Horizon Sensitivity
Longer horizon $\neq$ better planning



Why $H=1$ beats $H=5$:

MAE = 0.028 per step. At $H=5$, errors compound to 0.14 pH units — enough to recommend the wrong corrective direction. The planner optimizes a fictional future, not the real reactor.

$H=5$ achieves 0.9% in-target — worse than random (3.5%)

This is not a bug. It is the expected result when world model error is not negligible. Optimal horizon = $f(\text{accuracy} / \text{stochasticity})$, not a preference for planning depth.

Question 3 — What Should I Do?

100 fresh evaluation episodes. Same environment, same seeds. The planner used only the world model — no new reactor data.

Random Policy

3.5%

steps in target zone

Mean reward/step

-14.29

In-target rate

3.5% (35/1,000 steps)

Strategy

Uniform random



Model-Based (H=1)

11.3%

steps in target zone

Mean reward/step

-4.19

In-target rate

11.3% (113/1,000 steps)

Multiplier vs random

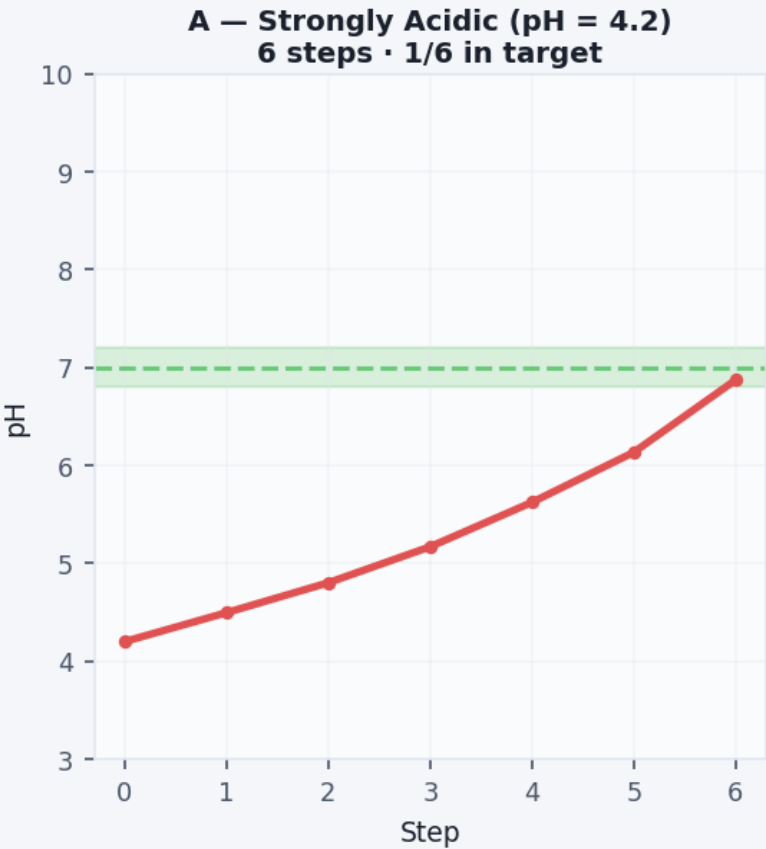
3.26×

Reward gain	Target rate	Multiplier	P(MB > Rand)
+10.10 per step	3.5% → 11.3%	3.26× vs random	100% bootstrap

The model never touched the reactor during training. The improvement is purely from offline learning and planning.

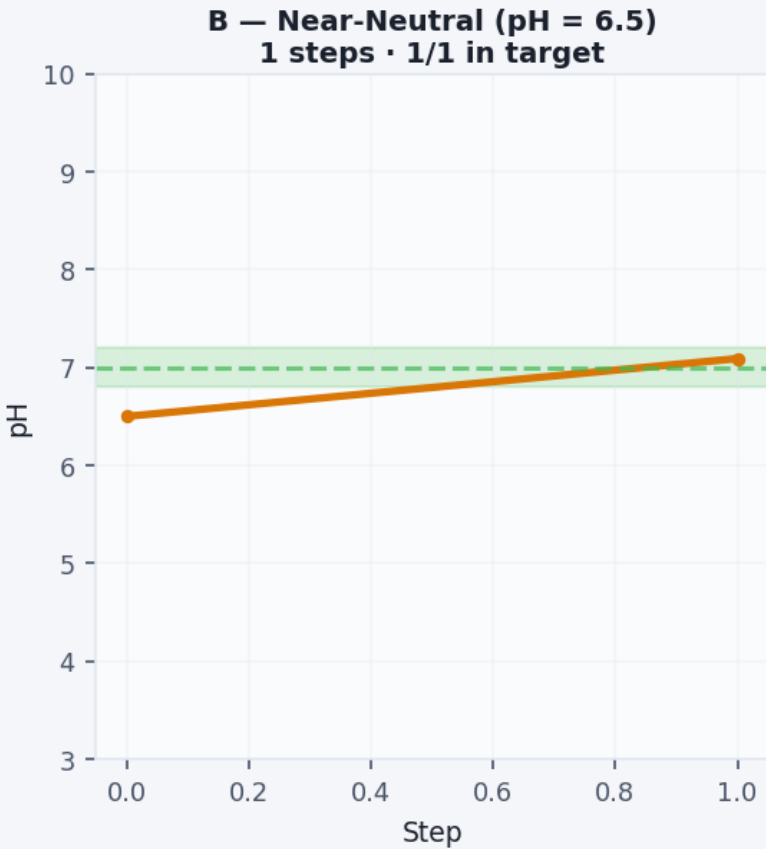
Three Reactor Scenarios — The Planner in Action

`control_ph(start_ph) → optimal dosing sequence until pH enters [6.8, 7.2].`



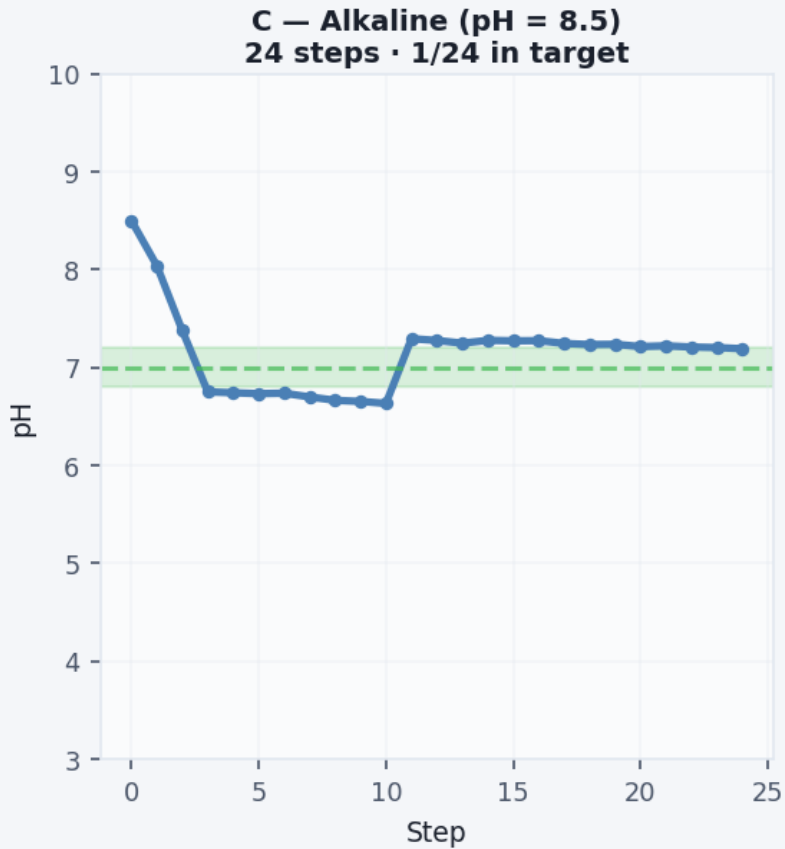
Scenario A — pH 4.2 · 6 steps

Applies Base +1.0 ml consistently — aggressive correction across 2.67 pH units. Switches to gentler dose as pH approaches 6.1.



Scenario B — pH 6.5 · 1 step

Single precise intervention: Base +0.5 ml pushes pH from 6.5 to 7.09. The planner at its most efficient — one decision, target reached.



Scenario C — pH 8.5 · 24 steps

Drops acid aggressively, then holds No Dose for 13 consecutive steps — waiting for natural drift to cross 6.8. Patience over over-correction.

Scenario C reveals the planner's planning: 13 steps of No Dose while waiting for drift — this is not reaction, it is foresight.

Five Lessons From the Finale

What model-based RL taught — and what it confirmed about everything that came before it.

01 The world model is the architecture.

$R^2 = 0.9987$ and $MAE = 0.028$ are not just metrics — they are the engineering budget that determines how far the planner can look ahead before reality diverges from simulation.

02 Longer horizon is compounded error, not deeper planning.

$H=5$ achieves 0.9% in-target — worse than random. Per-step error of 0.028 pH compounds to 0.14 units at $H=5$. The optimal horizon is $\text{accuracy} \div \text{stochasticity}$.

05 Model-based RL completes the portfolio's arc.

Supervised: learned to SEE patterns. Unsupervised: learned to DISCOVER structure. Model-free RL: learned to ACT. Model-based RL: learned to PLAN. Not the end — the question: what else can the model simulate?

03 Offline learning creates a coverage gap exactly where it matters.

41 training samples in the target zone [6.8–7.2]. The world model is least accurate where the controller operates most. Production requires active data collection.

04 The planner's decisions are operationally legible.

$\text{pH} < 6 \rightarrow \text{Base} + 1.0 \text{ ml}$. $\text{pH} \text{ near } 7.1 \rightarrow \text{No dose}$. $\text{pH } 7.4 \rightarrow \text{wait for drift}$. These emerge from the reward function through the world model — not from programming.

The Portfolio — 24 Industrial ML Projects

From logistic regression on labeled defects to a model-based planner that simulates reactor futures. One arc. One portfolio.



For twenty-three projects, the models observed and decided. This one plans.

The world model is the architecture. The planner is the intelligence. The controller is the result. All three together — trained offline, deployed in real time, operating without ever touching the reactor during learning — achieve 3.26× more time in the target zone than a random policy.

The arc of this portfolio mirrors the arc of the field: from observation to discovery to action to planning. Each step added a capability the previous one didn't have. The question now is not whether machine learning can serve operational excellence. The answer, across 24 industrial problems, is: yes — and in ways that classical statistics cannot reach.

Clone the repositories. Run the notebooks.

Change a reward function. Rebuild a world model. Extend to a new industrial problem.

Because Kaizen has no finish line — and neither does the portfolio.

